

Ground Control Witness Form

Project Overview: Lenton Lodge Gatehouse, Wollaton Park
Project Code: GATE

Date: Tuesday 21st November 2023

	Details	Notes/Comments	Justification/Explanation
Survey Overview			
Date:	Tuesday 21st November 2023	Date of survey.	Re-survey after 14 th November
Survey Conditions:	No rain expected. Visibility should be good.	Brief description of survey conditions	
No. of Points	8 Survey Markers (Thermal Targets) 12 Natural Markers	Number of points being taken.	8 thermal targets on central façade, with 6 additional to use as comparison. 6 to cover turrets
Equipment:			
Model:	Leica TS10 Total Station	Make/model of surveying equipment used. This includes the make/model of any TST, prism, tripod	
Specifications:	N/A	Critical specifications only inc. accuracy, range etc.	
Additional Equipment	2x Tripod, Prisms Tape Measure, Hammer, Nails	Equipment needed for the execution of the survey i.e. tape measure	Tape measure forgotten last time
Survey Team:			
Planned / Set Up by:	Neil Sutherland	Lead individual responsible for survey planning.	
Observed by:	Neil Sutherland	The individual taking the control point observations. This should remain consistent during the survey.	
Computed by:	Neil Sutherland Chenyu Xue Sean Ince	The team member responsible for the processing of the observations.	

Notes:	
Notes:	Survey of thermal targets and natural markers for Lenton Lodge Gatehouse, Wollaton Park This represents the critical control network for the PhD, with efforts prior to this being used as practice / scouting surveys. The 8 thermal targets will be surveyed, with an additional 12 natural markers being surveyed (see. Site Diagram 1). Points will be taken from 3 locations (TS001, TS002, TS003) and averaged out These serve several purposes: 1. To supplement the 8 thermal markers 2. To act as a means of comparison for designed and natural targets 3. To act as check points to test model accuracy 4. To provide control in areas where thermal targets could not be applied (notably turrets) Assistance is provided by Chenyu Xue. Naming protocol: TS001T1 – T8 TS001N1 – N12 TS002T1 – T8 TS002N1 – N12 TS003T1 – T8 TS003N1 – N12

Diagrams

Site Diagram 1:



POINT T1

	Details	Notes/Comments	Justification/Explanation
Point Name:	T1	Name saved within the equipment	Thermal 1
Easting (X): (m)	988.014		
Northing (Y): (m)	1021.547		
Height (Z): (m)	102.042		
CRS / Datum:	LOCAL		
Station Name/No.:			

Notes:	
Notes:	Bottom left thermal target, located by left door of the Lenton Lodge Gatehouse (when facing) Cross section of aluminium and rubber used as laser target

Diagrams

**Control Point
Photo(s):**



POINT T2

	Details	Notes/Comments	Justification/Explanation
Point Name:	T2	Name saved within the equipment	
Easting (X): (m)	986.726		
Northing (Y): (m)	1021.905		
Height (Z): (m)	106.079		
CRS / Datum:	LOCAL		
Station Name/No.:			

Notes:	
Notes:	1st floor left window thermal target, located above stone carving on the left side of the Lenton Lodge Gatehouse (when facing) Cross section of aluminium and rubber used as laser target

Diagrams

Control Point

Photo(s):



POINT T3

	Details	Notes/Comments	Justification/Explanation
Point Name:	T3	Name saved within the equipment	
Easting (X): (m)	986.697		
Northing (Y): (m)	1021.805		
Height (Z): (m)	108.691		
CRS / Datum:	LOCAL		
Station Name/No.:			

Notes:	
Notes:	2nd floor left window thermal target, located on stone ledge on the left side of the Lenton Lodge Gatehouse (when facing) Cross section of aluminium and rubber used as laser target

Diagrams

**Control Point
Photo(s):**



POINT T4

	Details	Notes/Comments	Justification/Explanation
Point Name:	T4	Name saved within the equipment	
Easting (X): (m)	988.000		
Northing (Y): (m)	1021.596		
Height (Z): (m)	112.058		
CRS / Datum:	LOCAL		
Station Name/No.:			

Notes:	
Notes:	Top left thermal target located on stone parapet (when facing Lenton Lodge Gatehouse)

Diagrams

**Control Point
Photo 1:**



POINT T5

	Details	Notes/Comments	Justification/Explanation
Point Name:	T5	Name saved within the equipment	
Easting (X): (m)	994.309		
Northing (Y): (m)	1020.190		
Height (Z): (m)	102.024		
CRS / Datum:	LOCAL		
Station Name/No.:			

Notes:	
Notes:	Bottom right thermal target, located by right door of the Lenton Lodge Gatehouse (when facing) Cross section of aluminium and rubber used as laser target

Diagrams

**Control Point
Photo 1:**



POINT T6

	Details	Notes/Comments	Justification/Explanation
Point Name:	T6	Name saved within the equipment	
Easting (X): (m)	995.671		
Northing (Y): (m)	1019.992		
Height (Z): (m)	106.034		
CRS / Datum:	LOCAL		
Station Name/No.:			

Notes:	
Notes:	1st floor right window thermal target, located above stone carving on the right side of the Lenton Lodge Gatehouse (when facing) Cross section of aluminium and rubber used as laser target

Diagrams

Control Point
Photo 1:



POINT T7

	Details	Notes/Comments	Justification/Explanation
Point Name:	T7	Name saved within the equipment	
Easting (X): (m)	995.658		
Northing (Y): (m)	1019.906		
Height (Z): (m)	108.672		
CRS / Datum:	LOCAL		
Station Name/No.:			

Notes:	
Notes:	2nd floor right window thermal target, located on stone ledge on the right side of the Lenton Lodge Gatehouse (when facing) Cross section of aluminium and rubber used as laser target

Diagrams

**Control Point
Photo 1:**



POINT T8

	Details	Notes/Comments	Justification/Explanation
Point Name:	T8	Name saved within the equipment	
Easting (X): (m)	994.419		
Northing (Y): (m)	1020.267		
Height (Z): (m)	112.016		
CRS / Datum:	LOCAL		
Station Name/No.:			

Notes:	
Notes:	Top right thermal target located on stone parapet (when facing Lenton Lodge Gatehouse)

Diagrams

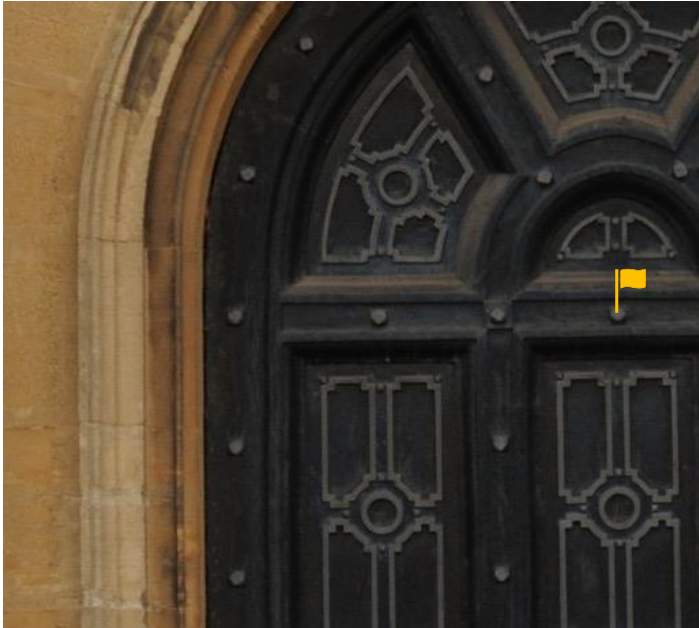
**Control Point
Photo 1:**



POINT N1

	Details	Notes/Comments	Justification/Explanation
Point Name:	N1	Name saved within the equipment	
Easting (X): (m)	986.749		
Northing (Y): (m)	1022.086		
Height (Z): (m)	103.486		
CRS / Datum:	LOCAL		
Station Name/No.:			

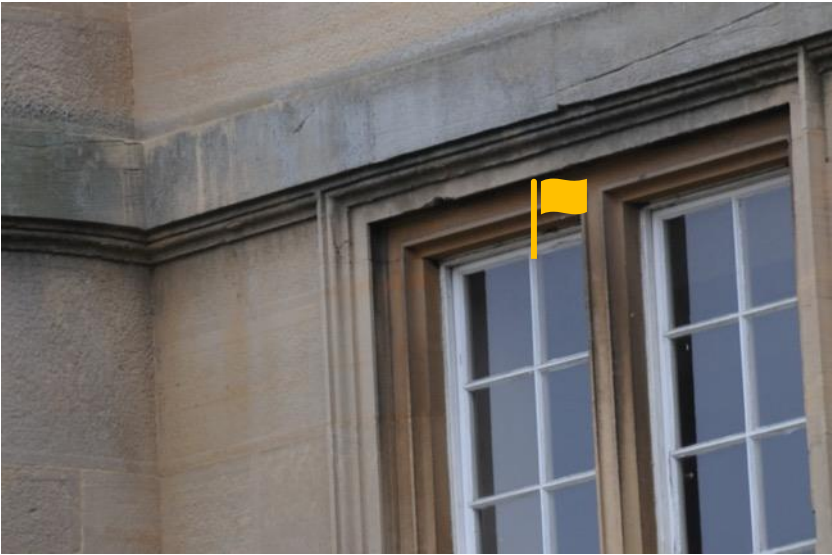
Notes:	
Notes:	Bottom left door, top line of 7 bolts, central nut under crescent of wooden door on left of the façade (when facing)

Diagrams	
Control Point Photo 1:	
Diagrams	
Control Point Photo 2:	

POINT N2

	Details	Notes/Comments	Justification/Explanation
Point Name:	N2	Name saved within the equipment	
Easting (X): (m)	986.420		
Northing (Y): (m)	1022.126		
Height (Z): (m)	107.511		
CRS / Datum:	LOCAL		
Station Name/No.:			

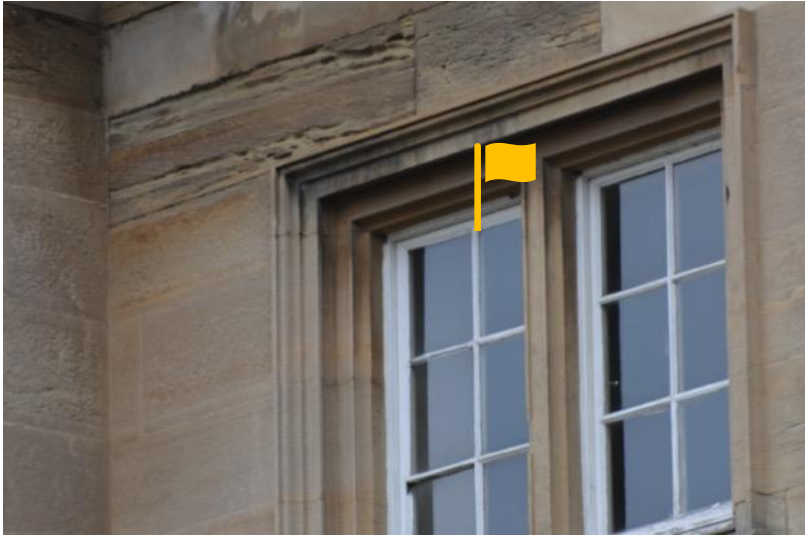
Notes:	
Notes:	Top left corner of left pane of middle window on left of the façade (when facing) Where glass meets the woodwork in the Grille on the left (on the inside corner of the pane)

Diagrams Control Point Photo 1:	
Diagrams Control Point Photo 2:	

POINT N3

	Details	Notes/Comments	Justification/Explanation
Point Name:	N3	Name saved within the equipment	
Easting (X): (m)	986.441		
Northing (Y): (m)	1022.146		
Height (Z): (m)	109.945		
CRS / Datum:	LOCAL		
Station Name/No.:			

Notes:	
Notes:	Top left corner of left pane of middle window on left of the façade (when facing) Where glass meets the woodwork in the Grille on the left (inside corner of pane)

Diagrams Control Point Photo 1:	
Diagrams Control Point Photo 2:	

POINT N4

	Details	Notes/Comments	Justification/Explanation
Point Name:	N4	Name saved within the equipment	
Easting (X): (m)	995.678		
Northing (Y): (m)	1020.17		
Height (Z): (m)	103.473		
CRS / Datum:	LOCAL		
Station Name/No.:			

Notes:	
Notes:	Bottom right door, top line of 7 bolts, central nut under crescent of wooden door on right of the façade (when facing)

Diagrams Control Point Photo 1:	
Diagrams Control Point Photo 2:	

POINT N5

	Details	Notes/Comments	Justification/Explanation
Point Name:	N5	Name saved within the equipment	
Easting (X): (m)	996.043		
Northing (Y): (m)	1020.076		
Height (Z): (m)	107.483		
CRS / Datum:	LOCAL		
Station Name/No.:			

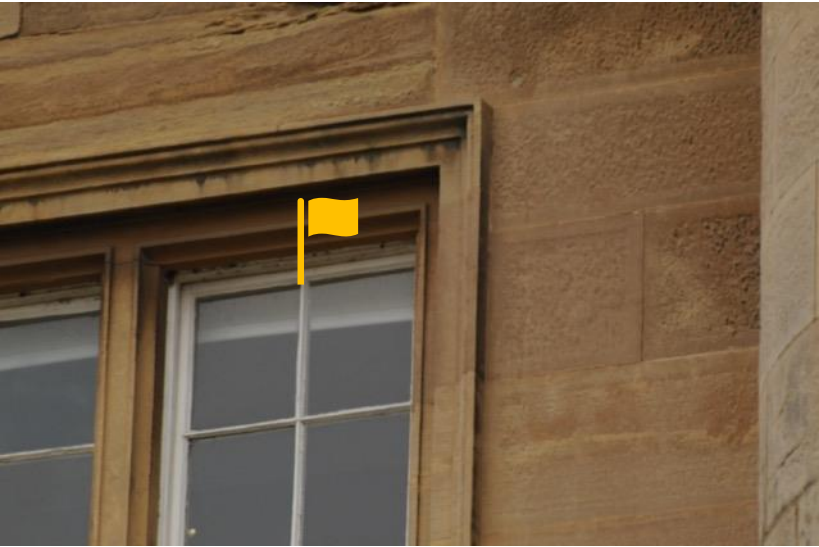
Notes:	
Notes:	Top right corner of right pane of middle window on right of the façade (when facing) Where glass meets the woodwork in the Grille on the right (on the inside corner of pane)

Diagrams Control Point Photo 1:	
Diagrams Control Point Photo 2:	

POINT N6

	Details	Notes/Comments	Justification/Explanation
Point Name:	N6	Name saved within the equipment	
Easting (X): (m)	996.063		
Northing (Y): (m)	1020.107		
Height (Z): (m)	109.938		
CRS / Datum:	LOCAL		
Station Name/No.:			

Notes:	
Notes:	Top right corner of right pane of middle window on right of the façade (when facing) Where glass meets the woodwork in the Grille on the right (inside corner of pane)

Diagrams Control Point Photo 1:	
Diagrams Control Point Photo 2:	

POINT N8

	Details	Notes/Comments	Justification/Explanation
Point Name:	N8	Name saved within the equipment	
Easting (X): (m)	983.927		
Northing (Y): (m)	1021.536		
Height (Z): (m)	109.972		
CRS / Datum:	LOCAL		
Station Name/No.:			

Notes:	
Notes:	Middle window of left turret, parallel to façade Point is the central grille of the window, inside corner where the pane meets the wood IMPORTANT - - - - - THERE IS NO N7!!!! - - - - - DUE TO LACK OF VISIBILITY IN DATASET, N7 WAS NOT INCLUDED!

Diagrams	
Control Point Photo 1:	
Diagrams	
Control Point Photo 2:	

POINT N9

	Details	Notes/Comments	Justification/Explanation
Point Name:	N9	Name saved within the equipment	
Easting (X): (m)	984.204		
Northing (Y): (m)	1022.334		
Height (Z): (m)	117.000		
CRS / Datum:	LOCAL		
Station Name/No.:			

Notes:	
Notes:	Top of left turret. Top baseplate connecting orb to turret, inside corner of square plate

Diagrams	
Control Point Photo 1:	
Diagrams	
Control Point Photo 2:	

POINT N11

	Details	Notes/Comments	Justification/Explanation
Point Name:	N11	Name saved within the equipment	
Easting (X): (m)	998.050		
Northing (Y): (m)	1018.520		
Height (Z): (m)	109.923		
CRS / Datum:	LOCAL		
Station Name/No.:			

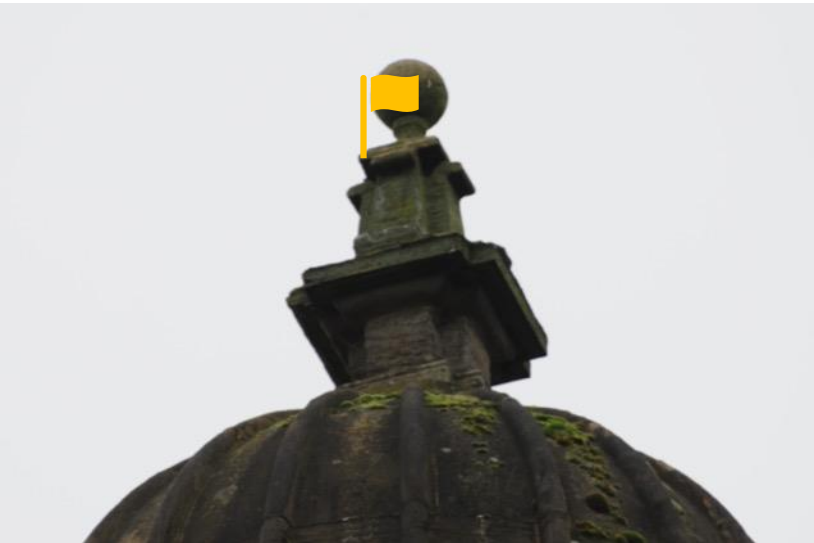
Notes:	
Notes:	Middle window of right turret, parallel to façade Point is the central grille of the window, inside corner where the pane meets the wood IMPORTANT - - - - - THERE IS NO N10!!!! - - - - - DUE TO LACK OF VISIBILITY IN DATASET, N10 WAS NOT INCLUDED!

Diagrams	
Control Point Photo 1:	
Diagrams	
Control Point Photo 2:	

POINT N12

	Details	Notes/Comments	Justification/Explanation
Point Name:	N12	Name saved within the equipment	
Easting (X): (m)	998.221		
Northing (Y): (m)	1019.394		
Height (Z): (m)	116.966		
CRS / Datum:			
Station Name/No.:			

Notes:	
Notes:	Top of left turret. Top baseplate connecting orb to turret, inside corner of square plate

Diagrams	
Control Point Photo 1:	
Diagrams	
Control Point Photo 2:	

Diagrams	
Additional Images:	